

Reviewer Pack — Minimal Hodge Decomposition Rank and Resolution Proof Width ...

Entry fa2807b81283 · INCONCLUSIVE

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Minimal Hodge Decomposition Rank and Resolution Proof Width Bound

Entry ID: fa2807b81283 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-07 07:59:49 UTC

1. Conjecture

Field A (mathematical branch): Hodge Theory **Field B** (complexity object): Boolean Satisfiability (Resolution Proof Complexity)

Statement:

For every Boolean satisfiability instance with resolution proof width w , the minimal rank of its associated Hodge decomposition is bounded by a polynomial function of w , specifically $|H(G)| \leq c(w)^3$ for some constant c and a finite-dimensional complex vector space G .

Rationale (proposer's reasoning):

The use of Hodge theory in complexity theory could potentially uncover novel structural insights into the resolution proof process. The minimal rank of the Hodge decomposition provides a geometric invariant that might correlate with combinatorial complexities like resolution proof width.

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 2affd587945da009

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, across all generated Boolean satisfiability instances up to $n=40$ variables, the mean of minimal Hodge decomposition ranks is less than or equal to the cube of the resolution proof width, with a p -value ≤ 0.05 from correlation analysis using 30 seeds.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 3 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "Hodge Theory" AND "Boolean Satisfiability" AND resolution proof complexity" - "minimal Hodge decomposition rank" AND "resolution proof width" AND polynomial bound" - "Hodge Theory" IN title AND "Boolean satisfiability" IN title AND "polynomial function"

Top relevant hits considered: - [http://arxiv.org/abs/1412.6019v1] The Hodge Theory of maps (Lectures 4 and 5) - [http://arxiv.org/abs/2005.07919v1] Hodge Theory of p -adic varieties: a survey - [http://arxiv.org/abs/1610.06246v1] The Hodge theory of the Hecke category

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=124, elapsed=240.4s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_instance(n):
        variables = [f'x{i}' for i in range(n)]
        clauses = []
        for _ in range(2**n):
            clause = random.sample(variables, k=random.randint(1, n))
            if random.choice([True, False]):
                clause = [f'~{var}' for var in clause]
            clauses.append(' '.join(clause) + ' 0')
        return '\n'.join(clauses)

    def dpll_solve(instance):
        # Simplified DPLL solver
        literals = instance.split()
        stack = []
        while literals:
            literal = literals.pop(0)
            if literal.startswith('~'):
                negated_literal = literal[1:]
                if negated_literal in stack:
                    stack.remove(negated_literal)
            else:
                return False
        else:
            return True
```

```

        stack.append(literal)
    return True

def hodge_decomposition(instance):
    # Placeholder for Hodge decomposition logic
    # This is a dummy implementation and should be replaced with actual computation
    return random.randint(1, 10) # Dummy rank value

n_max = 40
instances_tested = 30
total_rank = 0
resolution_widths = []

for _ in range(instances_tested):
    n = random.choice([5, 10, 15, 20, 30, 40])
    instance = generate_instance(n)
    width = dpll_solve(instance)
    rank = hodge_decomposition(instance)
    total_rank += rank
    resolution_widths.append(width)

mean_rank = total_rank / instances_tested
correlation = sum((rank - mean_rank) * (width - mean_width) for rank, width in zip(resolution_widths, total_rank)) / instances_tested
mean_width = sum(resolution_widths) / instances_tested

conjecture_holds = correlation >= 0.95 * math.sqrt(instances_tested)
counterexample = "" if conjecture_holds else "mapping_undefined"

return {
    "metric_name": "correlation",
    "metric_value": correlation,
    "instances_tested": instances_tested,
    "n_max": n_max,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47]
    results = []

    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_correlation = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_correlation} std=0.0 support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_correlation} std=0.0 support_fraction={support_fraction}")
    else:
        first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
        print(f"RESULT: FALSIFIED counterexample=\"mapping_undefined\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test timed out before producing data, which means the pre-registered support condition could not be unambiguously met. | next: NONE

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	13820
2	preregistration	ollama_remote	glm4:latest	0	0	9434
3	novelty	ollama_remote	glm4:latest	0	0	8655
4	novelty	ollama_remote	glm4:latest	0	0	9617
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	17588
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	45338
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13422
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11573
9	judge	ollama_remote	glm4:latest	0	0	52255

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 181704 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/fa2807b81283.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/fa2807b81283.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/fa2807b81283.tar.gz` (if generated)